

Documentation of external evaluation

Documentation of project evaluation by multiple, demonstrably qualified stakeholders and field experts is presented and is synthesized in a consistently specific, detailed, and thorough way; documentation is sufficient in two or more categories to yield meaningful analysis of that evaluation data; the synthesis of evaluations consistently addresses evaluators' specific questions, concerns, and opinions related to design requirements.

We had many days where we were visited by Dr Stephen Wancea and Dr. Heather Schwartz, along with scholars from vanderbilt's engineering program. With them, came multiple great ideas, along with major adjustments to our predecessors design of the project.

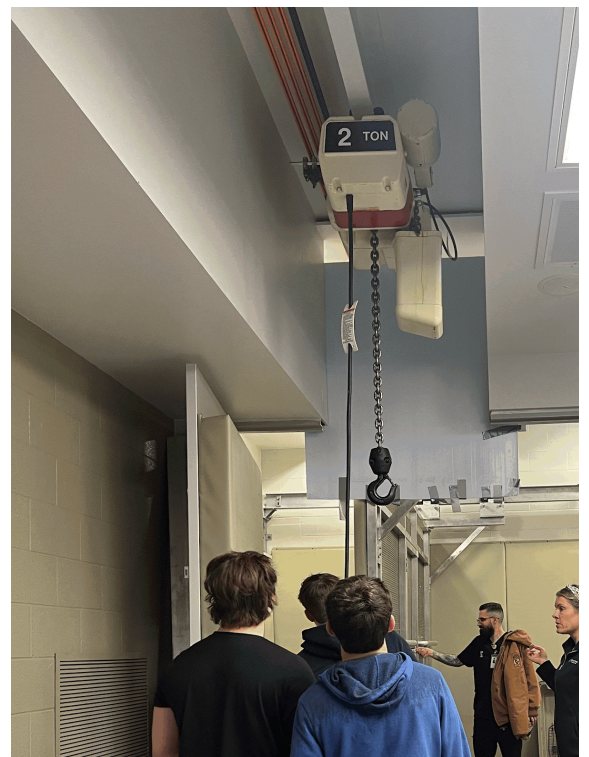
1. Stabilization

- The group before ours had a single engine leveler as their sole stabilizing function

As we analyzed the limitations of the previous design, our primary objective was to develop a solution that would allow us to stabilize the frame along both the horizontal and vertical axes. During one of our design review sessions, Bill Rodriguez—informally known as Bill-Rod and one of our dedicated volunteer overseers—joined us in brainstorming possible improvements. While we explored various options, Bill posed a critical question: was there room in the budget for a second engine leveler? Fortunately, the answer was yes. With this new opportunity, we decided to procure an additional engine leveler to mount alongside our existing one on the frame. However, while this addition addressed stabilization along one axis, it did not fully solve the problem—we still needed a reliable method to stabilize the frame along the other axis.

This is where Corbin, our head designer, applied his extensive hands-on knowledge and proposed a practical and cost-effective solution. He suggested integrating a chain block system—essentially a manual hoist—that could be used to raise or lower one side of the frame independently. This would allow us to counterbalance any shifts in the center of mass dynamically.

Traditionally, the automated hoist system installed in the rig had been used solely to lift and lower the suspended animal vertically. However, Corbin's proposal introduced a new application: by reconfiguring the hoist system to act on a single side of the frame, we could introduce a controlled pivot. This pivot would enable us to fine-tune the frame's orientation, both vertically and laterally, effectively allowing us to shift the center of mass in multiple directions. As illustrated in the image below, this innovation transformed our setup into a versatile, near-omnidirectional calibration system—precisely what was needed for achieving high-precision movement and stability during operation.



2. Fortifications

One of the biggest issues we ran into with our earlier prototypes was that the frame would warp or bend whenever any pressure was applied. It didn't take long to see that this was a major flaw—just handling the prototypes or adding components caused visible flexing. To get a better handle on the problem, we talked with a number of grad students and professors who came by to check out the early stages of our project. These conversations were super helpful. One of the grad students even suggested that we look into different support orientations—things like changing the frame's dimensions or rethinking the way it was reinforced.

We knew right away that we didn't want to mess with the frame's outer dimensions. Those were already dialed in to work with the other parts of our system, and changing them would have caused a domino effect of redesigns. So instead, we shifted our focus to finding better ways to support the inside of the frame. At first, we tried adding X-shaped cross braces to each corner. The idea was that diagonal supports might help distribute forces more evenly and reduce flex.

We built four different versions with variations on this X-bracing idea. While it did help a bit with the bending, it also introduced some new problems. The diagonals made it harder to assemble everything and limited our access to the inside of the frame, especially when we needed to mount or adjust components. So after some trial and error, we decided to switch gears.

Rather than stick with the diagonal approach, we started reinforcing the frame with more horizontal and vertical bars. This ended up being a much better solution. It

gave us the rigidity we were looking for, without making things harder to work with. It also made the overall design feel more modular and clean. In the end, this approach gave us a sturdier frame that was easier to build, easier to modify, and more reliable under pressure.